Jisang Ahn

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EDUCATION

University of Michigan **May 2026**
B.S. in Computer Science, High Distinction (GPA: 3.96/4.0) Ann Arbor, MI

- Courses: **Machine Learning**, **Reinforcement Learning**, **Computer Vision**, **Graphics & Generative Models**, Embedded Systems, Operating Systems, Compilers, Computer Organization, Networks, Databases, CS Theory, Deep Learning for Biology

EXPERIENCE

Software Engineer **June 2026 – Present**
Google Sunnyvale, CA

- First rotation in AI Catalyst program: agent framework for TPU workflows. Contributed to framework infra, agent runtime tools, etc. Enabled evals for agent-driven performance optimization on vLLM-TorchTPU and was a core contributor to the framework's sharding agent, capable of optimizing parallelism degrees, execution flags, etc. for both inference and training (MaxText) jobs.

Research Assistant (ML Systems) **Jan 2025 – May 2026**
Symbiotic Lab, University of Michigan Ann Arbor, MI

- Helped build a distributed runtime for LLM inference; wrote experiments to profile throughput vs. latency across model serving configs (e.g., disaggregated prefill/decode, varying TP degrees, per-executor batch sizes) for resulting paper submissions.
- Extended vLLM execution paths with additional disaggregation strategies for multimodal subsystems such as ViT, audio-encoders, and vocoders for various open-source models (e.g., Qwen3-Omni).
- For a different project, built a DNN energy usage profiler for Apple Silicon that works at the SoC-subcomponent level (for on-chip CPU, GPU, DRAM, NPU, etc.); integrated the library into Zeus, a PyTorch ecosystem research initiative for ML energy.

Software Engineer Intern **June 2025 – Sep 2025**
Amazon (Kindle) Seattle, WA

- Worked on Kindle embedded devices, building features in C++ for notetaking, book data, and networked cross-device sync.
- Engineered algorithms for cache reads/writes, asynchronous request-response cycles, and IPC handling; implemented extensive test automation and performance profiling.
- Optimized various disk & network bound operations by multithreading and ensured thread safety with mutexes.

Research Assistant (ML in Embedded Systems) **Oct 2023 – May 2024**
ISC Lab, University of Michigan Ann Arbor, MI

- Built an ultrasound classification system for real-time ML on edge devices (Orange/Raspberry Pi). Wrote Random Forest models in Sklearn, designed feature extraction scheme, and curated the dataset; yielded >84% accuracy across 10+ ultrasound activity classes.
- Optimized real-time inference for minimal latency; reduced response times substantially with an asynchronous, ring-buffer-based feature extraction pipeline (for resampling/filtering/encoding sensor inputs) with decoupled I/O & inference.

PUBLICATIONS

MBCE: TD-Guided Action Editing for Offline Model-Based RL.
Jisang Ahn. *Conference on Robot Learning (CoRL), Workshop on Modeling Uncertainty in Robotic World Models, 2026.*

Cornfigurator: Automated Planning for Any-to-Any Multimodal Model Serving.
Jeff J. Ma, Jae-Won Chung, Jisang Ahn, Yizhuo Liang, Runyu Lu, Akshay Jajoo, Myungjin Lee, Mosharaf Chowdhury. *Under submission, 2026.*

Cornserve: A Distributed Serving System for Any-to-Any Multimodal Models.
Jae-Won Chung, Jeff J. Ma, Jisang Ahn, Yizhuo Liang, Akshay Jajoo, Myungjin Lee, Mosharaf Chowdhury. *ACM Conference on AI Systems (CAIS), Demo Track, 2026.*

PROJECTS

PyTorch Extension for Transformer Attention (CUDA, C++, Python) **July 2025**

- Implemented FlashAttention as a CUDA kernel, packaged as a custom PyTorch extension accessible from Python
- Utilized strategies like online softmax and shared memory reuse, increasing GPU occupancy and throughput

GPU Accelerated Tiled Matrix Multiplier (CUDA, C++) **May 2025**

- Wrote CUDA kernels for tiled matrix multiplication, leveraging shared memory to minimize global memory access
- Optimized performance by parallelizing block-level computation and reducing memory latency bottlenecks

3D View Synthesis using Neural Radiance Fields (Python) **April 2026**

- Built a NeRF renderer in PyTorch that generates realistic images of a scene at arbitrary viewpoints, reconstructing a 3D scene from 2D images using an MLP with methods like sinusoidal position encoding and volumetric ray marching

SKILLS

- **Languages**: C, C++, Python, Rust, Java, C#, Swift, JavaScript/TypeScript, SQL, HTML, CSS
- **Tools**: PyTorch, JAX, NumPy, CUDA, vLLM, PostgreSQL, FastAPI, React, Docker, Kubernetes, NGINX, AWS, GCP